Clean Room Lab Report: Microfabrication of Thermocycler for PCR Applications

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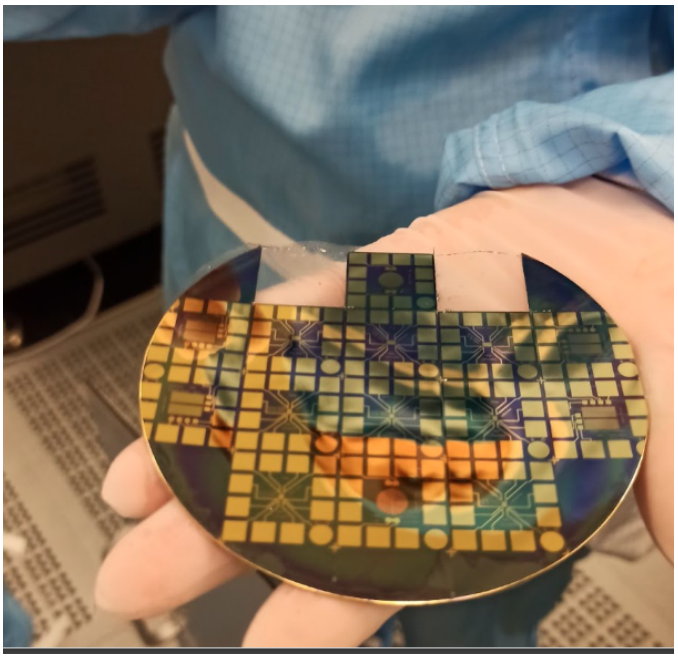


Fig. 1. PCT Thermocycler Chip Wafer

I. Context of the Chip

A. Overview of the Fabricated PCR Thermocycler Chip

This chip is a microfabricated chip designed for a PCR thermocycler. The chip instrument allows for control of thermal cycling, which is needed for DNA amplification in the polymerase chain reaction (PCR) process. It allows for PCR processes to be performed with miniaturized point-of-care diagnostics and lab-on-a-chip or lab-on-a-device devices, where speed and volume are critical. The chip provides thermal cycling at the microscale. The chip has structures of heater resistors, temperature sensors, and conductive pathways, all of which were patterned through photolithography for high accuracy and reproducibility at the microscale.

B. Background and Applications of PCR Technology

The polymerase chain reaction (PCR), which will be described in the entries below, is a method for amplifying DNA sequences. It uses temperature-dependent enzymatic reactions to selectively replicate DNA sequences. Amplified DNA has many applications in a variety of fields including clinical diagnosis, forensic medicine, and viral detection including COVID-19 testing. PCR has the ability to detect and analyze small quantities of DNA, making it a powerful tool in medicine and research. Typical benchtop thermocyclers are bulky, are high power, and not field-deployable, while this chip allows a more miniaturized, much efficient, and more field-deployable method of performing PCR. Miniaturized thermocyclers require precise thermal control, which is achieved through clean room microfabrication technology.

C. Thermal Architecture and Photolithographic Definition

During PCR, microscale levels of heating and temperature measurement are needed, making metallic thin films patterned into resistive elements that create heat via Joule heating ideal. A second resistive element acts as the temperature sensor and can monitor the temperature of the reaction in real time. The shape and location of the resistors are determined before patterning, using photolithography. Micro-scale precision is achievable only with clean room practices in which contamination and particles are minimized.

D. Advantages of MEMS-based PCR Devices

Utilizing MEMS (micro-electromechanical systems) technology allows for the production of micro-sized devices. The technology also provides faster heating-cooling cycles, reduced energy usage, and portability. In addition, the less consumption of reagents, and less waste produced is more environmentally friendly. These features are especially advantageous for portable diagnostic platforms and point-of-care considerations.

E. Overview and Objectives of the Laboratory Work

Through this lab, students experienced each step of the fabrication process as a reinforcing activity between knowledge and practice. The purpose of this laboratory work was to create a micro PCR chip. The necessary steps for fabrication involved clean room procedures, metal deposition, photolithography, etching, and characterization, each performed with extreme detail. These fabrication steps were critical to demonstrating

how the features and functionality of the final microdevice were affected. The overall goal of the project was to design and fabricate a silicon-based thermal cycler chip using standard clean room processes, which could then be characterized for its thermal and electrical properties in order to assess its ability to be integrated into micro-scale PCR. The chip produced demonstrates the technical precision, and challenges, associated with microfabrication workflows. Projects like this expose undergraduate students to the fundamental nature of the interdisciplinary MEMS device fabrication process.

II. Methodology

A. Pre-fabrication Procedures

Because of the size range of a device fabricated with microtechnology processes, around a few micrometers for a typical layer, which is around the diameter of a spore or a bacteria, there is a protocol before entering the clean room to limit contaminants transported by a human body, from hair to clothing dust, as well as outside organisms like virus. As some steps of an integrated circuit process like wet etching and photolithography need a room temperature of 20°, we deposit our bags, coats and sweats in a locker. Before anything, if needed, we could go to the toilets, as our clean room lab lasts 4 hours. The cloakroom is divided in two halves with two sticky surfaces at each side, separated by a low metal platform, used to sit on. First, we need to clean our hands with a hydroalcoholic solution and walk through a sticky surface to remove the maximum amount of dirt. Then, we need to wear a plastic cap, gloves and shoes protection, ensuring our shoes do not touch the next sticky surface. So, beginning with the foot near the next half of the room, we then equip the other foot with the aforementioned protection. Next, we wear a lab suit, and boots on top of the shoes protection. Finally, we could enter the lab.

B. Wafer fabrication and Standard Cleaning

Wafers are fabricated using the Czochralski process, where the silicon is melted into bars then cut into round slices before being doped. There are 4 types of wafers : 100N-type, 100P-type, 111P-type and 111N-type. [1, 33] The names of the different wafers is due to the orientation of the tilling. After its fabrication, wafers are cleaned using acids. Standard cleaning is critical to reduce defects and remove contaminants from the wafer. For silicon wafers, this step follows a 1-min bath of hydrofluoric acid then a bath lasting 3 min into water. Then, a high temperature dive into sulfuric acid for 3 min at 150° followed by a 3-min bath into nitric acid. Finally, a 3-min bath of deionized water then a 1-min dive into hydrofluoric acid again ends this cleaning step.

C. Oxidation

Silicon thermal oxidation is a high temperature process with temperatures ranging from 850 to 1150 ° C, needed to fabricate an insulating layer or diffusion barrier for dopants. As oxidation deprived the silicon, around 46% of the obtained silicon dioxide SiO insulating layer comes from the initial

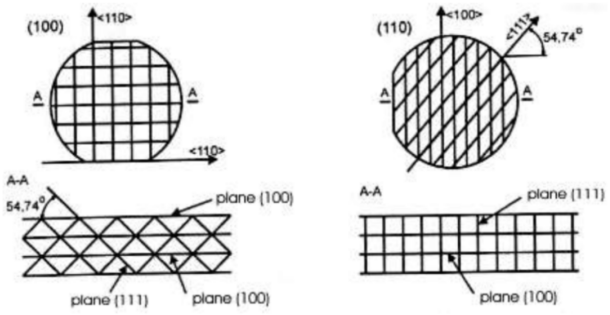


Fig. 2. Wafer Types



Fig. 3. Furnaces for Oxidation Step

silicon thickness. The wafer is put into a furnace where gas is pumped around it. There are two types of this process : dry and wet. The main difference is the usage of vapor instead of dioxygen. The chemical reaction is shown below :

$$\mathrm{Si} + \mathrm{O_2} \longrightarrow \mathrm{SiO_2}$$

$$\mathrm{Si} + 2\,\mathrm{H_2O} \longrightarrow \mathrm{SiO_2} + 2\,\mathrm{H_2}$$

There is a compromise between the oxidation time, the temperature of the process and the thickness obtained. The Deal and Groove model could be used to estimate the resulting thickness, we could see in this diagram that the maximum dry oxidation is around 0.6 µm whereas the maximum wet oxidation is around 3 µm for a 10 hour oxidation process. [3, 12]

D. Metal Deposition

Metal deposition is the 1st step in our clean room lab, as the previous steps have been done for us. Sputtering deposition is

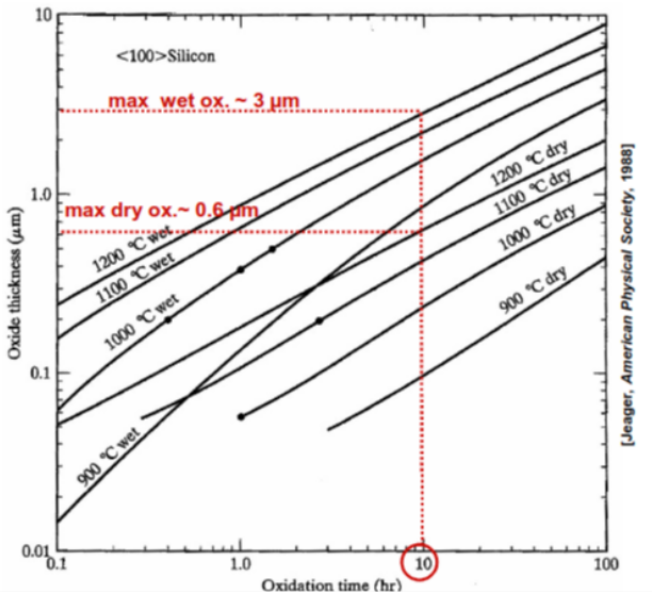


Fig. 4. Oxidation Thickness



Fig. 5. Metal Deposition Setup and Buttons

using a plasma to accelerate positive ions onto the target metal and ejecting atoms from it into the wafer.

The processed steps are :

I. Open the wafer holder, put the wafer upside down. On the tablet controlling the machine, press “another wafer”.

II. Press the following buttons on the board :

1) Pompage : Pumping air to create a vacuum.
2) (Vent SAS) : Only when opening the wafer holder.
3) Géné DC : Starting generator.
4) Transfert fermé : Enabling a mini-chamber to charge the wafer in chambers of different pressure.
5) Cache : Enabling the cache on top of the metal.

III. On the right panel with the different metal options, select your metal.

Note that each step may need some wait, like the injection of air where 1-2 min are needed to lower the pressure at atmospheric levels.

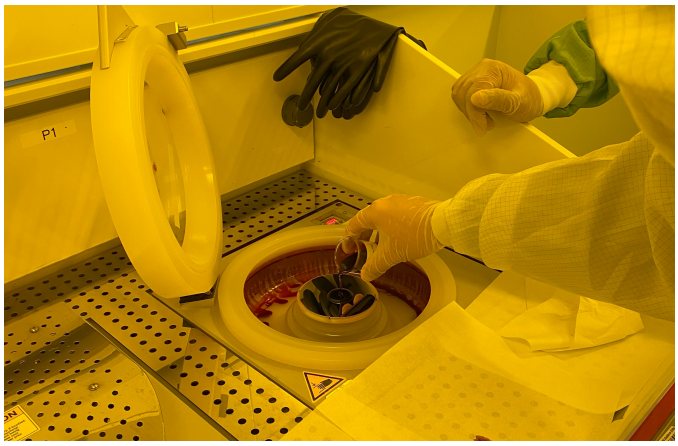


Fig. 6. Spin coater and photoresist beaker

E. Primary Photolithography

Microfabrication requires the transfer of layer designs onto the previously deposited metal layers on the silicon and glass wafers. This step is executed through the process of photolithography which can be defined as the utilisation of light to transfer geometric patterns on a light-sensitive chemical layer [1]. This process demands a tuned light source, a mask with inscribed layer design, and the photoresist, the mentioned light-sensitive chemical. Each element allows variation in layer design quality. In mask production, opaque and transparent zones can be used for defining either conductive or protective areas respectively. The polarity of the photoresist dictates which zones dissolve or reside. In positive photoresist, opaque areas in the mask remain and the transparent areas are dissolved. In negative photoresist, opaque areas are etched and transparent zones reside [1] .

In production of the thermo-cycler, two metal layers are deposited to act as the conductors. The heat produced is determined by the resistance of the main resistor and the current source. To control temperature, current and power should be related which is only possible with a known and fixed resistance.

$$R = \rho \cdot \frac{L}{A}$$

Resistance depends on the resistivity of the material, cross-sectional area of the conducting metal layer and length of the resistor. The primary photolithography enables the fabrication

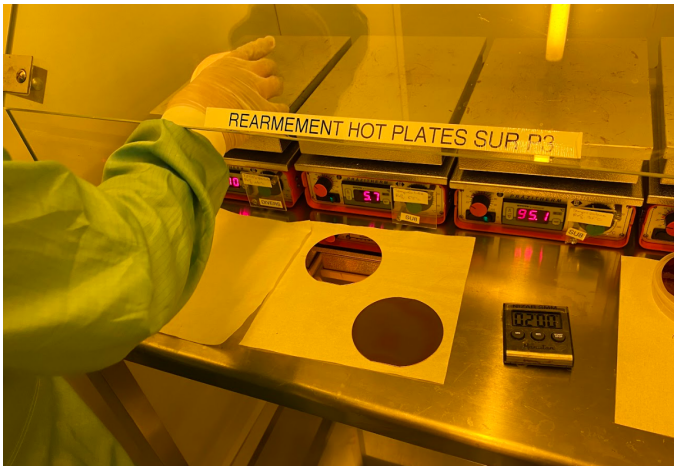


Fig. 7. Hot Plate Setup for Photolithography Steps

of the resistors with a precise topography that elongates the main resistor, allows the measuring resistor to be near the main resistor but not in contact, and also has a known resistance for temperature control. Material selection controls ρ, and circuit topography controls L and A.

The photolithography process is executed in a special room with monochromatic yellow light. The reason behind this light is to eliminate accidental exposure for the photoresist. Hence, wavelengths under 500 nm are blocked [2]. First, upon observation, the mask electrode is determined that it defines the conductive zones with opaque and the protective with transparent areas. Hence, positive photoresist is selected to be PFR 7790G-27cP and its layer thickness to be 1.1 μm. Uniform coating and thickness of the photoresist is controlled in a spin coater. SÜSS MicroTec spin coater is used with the parameters of 3125 speed, 500 acceleration and 30 seconds time. However before placement, the crimson colored photoresist is taken to a small beaker and spin coater is cleansed with acetone. Before placing the wafer onto the spin coater, an nitrogen gun is used to dust off any contamination. The placement is done by special tweezers that allow the correct alignment with the placement pins. After securing, the photoresist is poured in the center. The center alignment is crucial as well as the coarse eye-based quantity of the photoresist on the wafer. If not properly executed, insufficient coating could cause defects in design implementation. During this, the scientist should be careful about the hot plates that will be used after the coating. Thus, to prevent any delays in initiation of hot plates, they are checked and turned on. Then, the spin coater's lid is closed and it is started with the aforementioned settings. In this setting, only one speed step was used.

When the chromatic red layer is seen in each section of the wafer, they are taken to the 110 celsius hot plate where a 2 minute timer was started as well. After that, the wafer was left to cool and relax above tissues for 5 minutes. During unloading the hot plate, the tweezers were used to bring the wafers to the edge of the plate, and then grasped to obtain minimum contaminating contact.

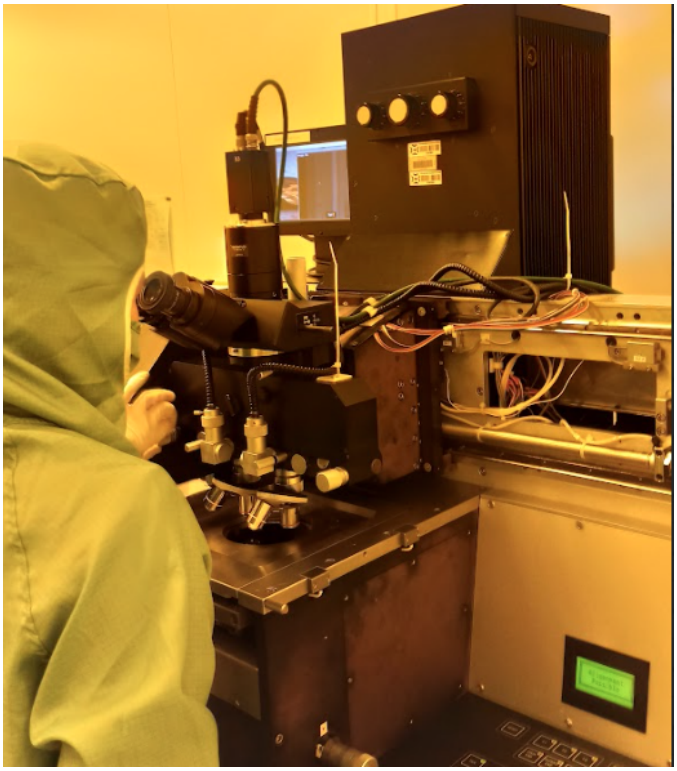


Fig. 8. Photolithography UV Exposure and Alignment Setup

Soft bake is followed by the most essential step in which substrate wafer is aligned with the electrode mask and then ultraviolet light source settings are configured for the design imprintment. For this procedure K&W MA 1006 machine is utilised where at first, the mask compartment is opened like a lid and the mask placed such that its shining grey surface is at the bottom. Then, as the mask lid is closed, a sliding substrate tray is taken forward. This tray is pressurized for holding the substrate in its place. When the tray fully slides, the pressure is turned off and wafer placement is enabled. At this stage, the bottom line is aligned with the tray screws inside the red circle. As the tray slides back into position, The microscope embedded on top slides with a rotor and allows the checking and alignment process. As this was the primary photolithography, only alignment was the correct positioning of the substrate in the tray and checking the circular boundaries. Then, the required settings are chosen which are as follows:

TABLE I
EXPOSURE PARAMETERS

Parameter	**Value**
Exposure Type	HARD
Time	3.5 s
Distance	100 μm
Power Density P	5.5 mW/cm^2

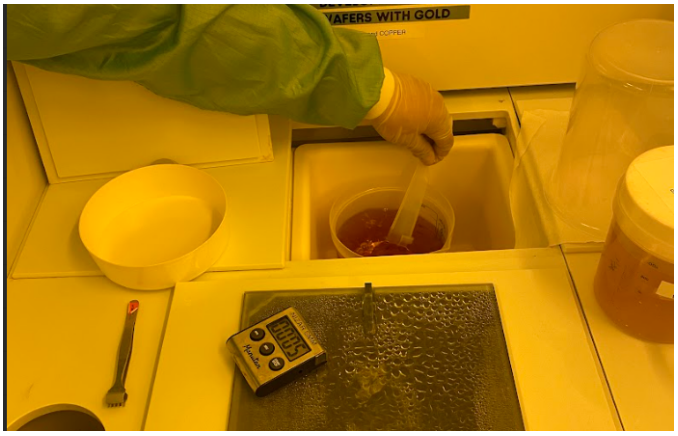


Fig. 9. Primary Photolithography Developer Stage

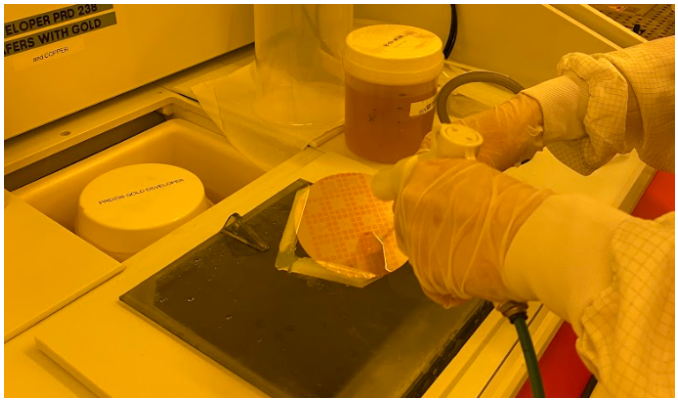


Fig. 10. Primary Photolithography Nitrogen Gun Cleaning

With these settings, the automatic UV light exposure sequence starts. In order to achieve anisotropy, the mask and the substrate should be close and in contact. However, to balance the deviation of light beam pathways due to lens properties, the distance is configured for precision. Exposure type indicates the contact type. In the sequence, the UV light source slides on top of the mask and substrate. After the assigned duration, it closes off and slides back into original position.

During UV exposure, the photoresist undergoes a chemical reaction and then these reacted areas can be solved using a chemical agent called developer. In this step, the wafer is placed inside a white plastic holder and secured with a hinge, and then the substrate is submerged inside MF-26 Developer. The wafer is constantly waved and turned to improve development. With a timer, 50 seconds were measured and then the wafer was cleansed with deionized water in a separate container. With constant bubble flow, the dissolved photoresist is swept off.

The photoresist layer after the hard bake needs to be measured and observed before the metal etching to minimize the waste in case of a systematical error in implementation. Under the Keyence VHX optical microscope, the layer is observed, the width of the connections and the main resistance is measured. This part of the production sequence will be explained further in characterisation section.

To fully complete the evaluation, the thickness of the layers needs to be measured and microscope is insufficient for this purpose, Hence, a Bruker DektakXT computer-controlled profilometry device is utilized, where the wafer is placed under a measuring needle. As the distance of the needle is at first given to the system, the starting point and end point for the small incision-like movement is defined. The needle mechanism gives the differential length and as we start from a protective layer zone to a conducting zone, the thickness increases. In the profilometry result, the thickness is measured to be 1.1159 μm which has 1.44 % percentage error that is in the acceptable tolerance range.

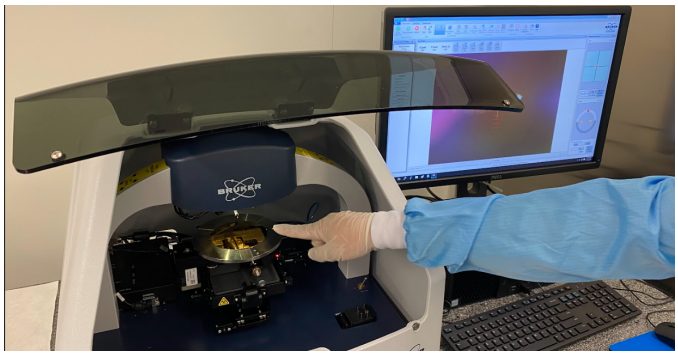


Fig. 11. Profilometry Setup

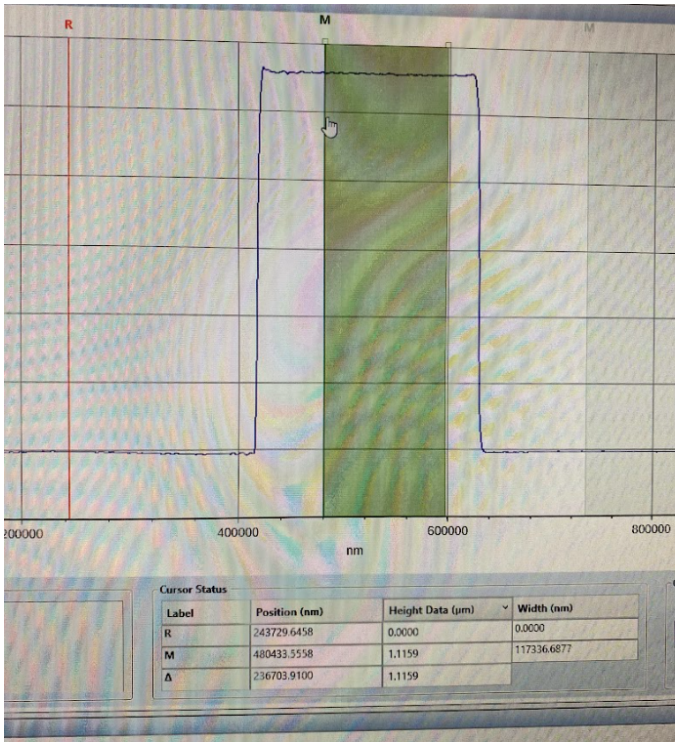


Fig. 12. Primary Photolithography Profilometry Result

F. Metal Etching

1) Purpose: Metal Etching is a process that is required to isolate the regions where metal layers are desired and to remove other areas where metals are not required. After the photolithography step, photoresists are applied on the desired areas, leaving the rest exposed to define the etching areas. This is done to create conductors like heater or sensor elements. After this step, the framework of conductors of the circuits is created. Without these steps, there would be undesired metal residues which would affect the circuits in some ways such as short circuits, resistance irregularities, and aesthetic deterioration.

2) Materials and Chemicals : This process involves etching two distinct metal layers: Gold (Au) and a Titanium-Tungsten (TiW) alloy. Gold is a noble metal that is mostly unreactive to chemicals. A specific solution called KI-I2 is being used to selectively dissolve undesired gold residues. KI-I2 is a special solution that is used to specifically wash away noble metals. High concentrated KI-I2 can solve gold at room temperature,slowly but in a controllable way. The solution contains iodine (I) and potassium iodide (KI), which together form I ions that actively dissolve gold. *Aqua Regia* is an extremely aggressive, non-specific and very abrasive solution. Also, cyanide-based etchants are very effective but very toxic and therefore prohibited in most countries. This solution specifically etches noble metals while having minimal effects on other metals. This increases the selectivity of the process, preventing undesired losses. Titanium-Tungsten complex (TiW) is a resistive layer to bind gold to SiO. Hydrogen Peroxide (HO) is used to etch this layer. However, it is hard to etch a resistive metal like TiW. Therefore, ultrasonic agitation is applied to enhance the etching efficiency. A very critical step during this process is timing, if metals stay for too long in the solution it causes an over-etching, which means the design is corrupted. In contrast, with insufficient time, it means under-etching. Which means fail to etch completely.

Reaction for Gold Etching:

$$\mathrm{Au} + 2\,\mathrm{I_2} + 2\,\mathrm{I^-} \longrightarrow \mathrm{AuI_4^-}$$

Reaction for Ti Etching:

$$\mathrm{Ti} + 3\,\mathrm{H_2O_2} \longrightarrow \mathrm{TiO_2} + 3\,\mathrm{H_2O}$$

Reaction for W Etching (More passive than Ti):

$$\mathrm{W} + 3\,\mathrm{H_2O_2} \longrightarrow \mathrm{WO_3} + 3\,\mathrm{H_2O}$$

3) Environment and Safety: These steps must be conducted with strict adherence to safety protocols due to the use of hazardous chemicals since very reactive and dangerous chemicals are used during these steps. To prevent as much risk of accidents as much as possible, these actions are taken in specific wet etching rooms. Cross-contamination with other metallic processes can compromise the integrity of the wafer, potentially leading to complete device failure. Specific glasses, gloves and aprons are needed in this room. Due to volatile chemicals, filtered air-conditioning is needed. In an ultrasonic bath, there is a risk of overflow. It requires intense attention.

4) Procedure - Step-by-step: Initially, the gold layer is selectively removed using a chemical etchant, which is a complex solution called K1-I2. An exposure time of approximately 1 minute is sufficient to etch a 100 nm thick gold layer using KI-I2 solution. This process requires strict safety precautions. Absorbent pads soaked in deionized water are placed around the etching bath to mitigate the risk of chemical spills or overflow. It is critical to ensure that all of the wafer has been introduced to the solution equally during the process. Subsequently, the underlying Titanium-Tungsten (TiW) layer is etched using hydrogen peroxide (HO) under ultrasonic agitation for 30 to 45 seconds. HO used as an active chemical (oxidising agent), ultrasonication is needed for a rapid process and also for a homogeneous etching. The timing is proportional to the thickness of the wafer. Throughout the process, photoresist acts as a protector from etching agents, which is designed in the previous step, photolithography. After etching, this layer will be removed in the subsequent photoresist stripping step. These steps ensure a highly selective and controlled etching process.

5) Outcome: Patterned gold and TiW layers remain only in required regions, afterwards, an visual inspection is performed, also an inspection could be done with a Keyence VHX optical microscope in order to assess the quality of the patterned features, including line width and edge definition. If there are no residues or overetches, the process has been done successfully and ready for the next step: photoresist stripping.

G. Photoresist Stripping

1) Purpose: Photoresist is a light-sensitive polymer layer used to either protect or expose specific regions during photolithographic patterning depending on its type. After the metal etching process, this layer loses its function and causes problems if it remains on the surface. These problems include causing chemical contamination before a new process, disorders of patterning in new photolithography steps, and also probability of causing a problem in electrical conductivity which is the main part of this device. This surface must be cleaned perfectly. The cleaning is sensitive, therefore both chemical and mechanical methods are used at the same time.

2) Material and Equipment: There are multiple chemicals which are used to complete the cleaning. Acetone is the strongest solvent in the process. It rapidly dissolves organic residues of photoresist. Its high volatility also helps in quick evaporation after application. However, it is not enough by itself since after the process some residues can stay still. For the second step, Ethanol or IPA(Isopropyl Alcohol) is used. It clears the residues after acetone and greasy trails. It is a gentle solvent compared to acetone. At the end, Deionased water (DI Water) is used to get rid of any of the chemicals and also from leftover ions, ensuring a completely clean surface. Usage of DI water prevents mineral deposits that may result from tap water. To dry water, a nitrogen gun is used. Nitrogen is preferred over air to avoid oxidation risks. No water leftovers

should stay after the blowing of the nitrogen gun. This process requires attention while being sure no droplets are existing. In the final step, the wafer is placed in an oven at 250°C for 15 minutes to ensure complete drying. This step also improves surface stability before the next lithography layer which will be placed in next steps. Solvent purity is also very critical since contaminants can be re-deposited during rinsing.

3) Environment and Safety: Since the process involves volatile chemicals, steps are taken in fume hoods to ensure proper ventilation. This prevents inhalation of harmful vapors and minimizes explosion risk. Gloves, glasses and lab coats are compulsory. These protective gears prevent direct exposure to chemicals and thermal hazards. Acetone and IPA are highly volatile solvents with significant flammability risks. Open flames or heat sources must be strictly avoided nearby. The drying oven operates at high temperatures and therefore it requires strict attention during use. The nitrogen gun operates under high pressure and must be handled with care to avoid accidental discharge or wafer damage. Direct contact with the wafer should be minimized to prevent delamination.

4) Procedure - Step-by-step: The wafer is placed on a spin coater carefully and fixed there. Tweezers are used to avoid contamination or damage. Tweezers are used on blind areas where there are no metal or resistor layers. After that acetone is spread all across the wafer homogeneously. Acetone dissolves the majority of the photoresist layer. Also, this rotational movement helps it to spread equally. Thanks to centrifugal force, liquid moves radially outward. After a while, ethanol is dropped onto it in the same manner. After the solvents have fully interacted with the surface, the wafer is rinsed with deionized (DI) water. Then, drying with a nitrogen gun is done to complete the process with an oven, 250°C for 15 minutes. This final bake step removes moisture and stabilizes the surface. The spin coater creates a centrifugal force that enables uniform chemical distribution, ensuring homogenous cleaning. It also prevents pooling or streaking of the solvents. Each solvent is applied sequentially without allowing the previous one to fully evaporate, maintaining surface wetness.

5) Outcome: In a successful stripping process, the surface of the wafer should be clean and bright. The visual cue often indicates the absence of organic residue. While examining under the microscope there should be no photoresist residues. Once the stripping process is completed successfully, the wafer is ready for a new process like SU-8 photolithography. The presence of residue at this stage may interfere with SU-8 adhesion. If there are still residues, it should be known that new additional stripping steps can be required but more strippings can cause over-abrasion in wafer. Repeated cleaning may damage the wafer surface.

H. Secondary Photolithography

After photoresist stripping, the last major step of the manufacturing is the passivation. To prevent unwanted contact, only the connection pins needs to be opened. Hence, the remaining parts require an insulation layer. For this purpose, a secondary photolithography step is performed to define the insulated zones and uncovered contact areas. Insulation layer material is chosen to be SU8 2002 that is a negative photoresist.

First, under monochromatic yellow light, the wafers are cleansed with a nitrogen gun. To obtain the desired layer thickness, SÜSS MicroTec spin coater is prepared. The photoresist is observed to be transparent, hence to prevent any future contamination, the inside surfaces of the spin coater is wrapped with aluminum foil. Through the usage of special tweezers, the wafer is placed on the rotating center, while the photoresist is poured at the center with a small beaker. The amount of the photoresist is checked to eliminate insufficient thickness. Spin coater is programmed to have two sequential spin parameters: First, 500 rpm speed, 200 rpm/s acceleration, and 5 seconds spin time. Second programme with 3000 rpm speed, 300 rpm/s acceleration, and 30 seconds spin time.

For the soft bake stage, the wafers are taken to a pre-started hot plate that is adjusted for 65C. Silicon wafers are timed for 1 minute . After that, with utmost care given upon holding the wafers, they are placed on top of another hot plate for 95 C. At this temperature, wafers are held for 1 minute as well. For the last soft bake step, wafers are placed to 65C hot plate for 1 minute. The substrates are left for 5 minutes of relaxation period. Even though, all wafers in this lab session were silicon, the glass substrates would have required for 2 minute at the 65C and 4 minutes for 95 C during soft bake.

For UV exposure, the wafer placement is executed like described in the primary photolithography section. Passivation mask is placed. The substrate tray is extended fully and wafer is placed with attention to placement circle and indicator screws. At this step, the alignment is carefully executed through K &W MA 1006 mask aligner. Alignment markers which are lock-and-key based plus markers that conforming plus patterns in different sizes are aligned with their counterparts on the mask. If not properly aligned, passivation could block connection ports or expose resistive sections. Through a microscope, the patterns are matched with three knobs; vertical, horizontal and rotational. this alignment can be seen in figure . Then, UV source is adjusted and the parameters in table IV are selected.

TABLE II
EXPOSURE PARAMETERS

Parameter	**Value**
Exposure Type	SOFT
Time	15 s
Distance	150 μm
Power Density P	5.5 mW/cm^2

As SU8 is a negative photoresist, the opaque masked areas are made to be soluble where light-exposed regions are hardened. ”SOFT” mode determines the contact mode while the other parameters are specific fro the photoresist.

The exposed wafers are then taken to hot plates. Sequentially, they are heated for 1 minute at 65C, 2 minutes at 95 C, and 1 minute at 65C again. The wafers are then left to cool down for 10 minutes. These values are measured with

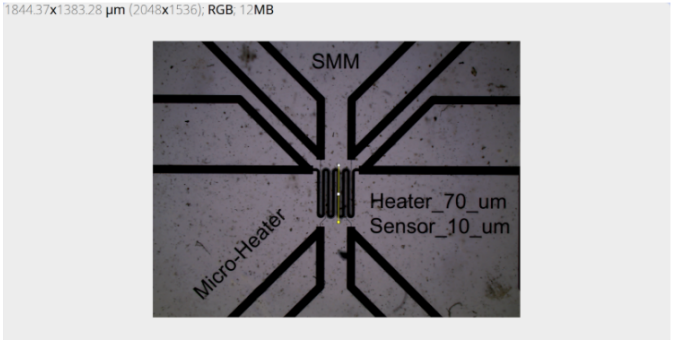


Fig. 13. Microscope View of the Chip

timers and if a glass substrate was used in this lab session, the timings would be 65C/ 2 minute - 95 C/5 minutes - 65C/ 2 minute.

Similar to the primary photolithography step, the wafers are submerged into developer chemical within a peitri dish. The dish placed on a rotater such that homogeneity is accomplished while 3 minutes timers are started. Then, the substrates are taken and rinsed in IPA and then deionized water.

Hard bake for the secondary photolithography is not completed in our session due to time limitations, however SU8 requires 30 minutes of heat treatment at 175C for completion of the microfabrication. An optical microscope is used to evaluate on the removal of the photoresist on the contact ports.

Profilometry is performed using a Bruker DektakXT system for thickness measurement. A fine needle is scanned across a defined region, measuring step height between protected and exposed areas. The thickness measured to be 1.2650 μm.

III. Characterisation

A. Description of the Chip with Video Microscope

Using ImageJ, an image processing and analysis program, we measured the width and the height by crossing a line between two unicolor areas, using a scale of 4.338μm/pixel with a color camera parameter MAG, of 2. The inner resistance Rm has a MAG parameter of 20, so it's about 13μm. The measurements can be seen in the corresponding figures. The heating resistor is found to have a width of 190μm and a height of 235μm.

Given the dimensions of the heating resistor observed under the video microscope:

- Width $W = 190\,\mu\text{m}$
- Length $L = 235\,\mu\text{m}$
- Measured resistance $R = 200\,\Omega$

The sheet resistance $R_\square$ is calculated using the formula:

$$R_\square = \frac{W}{L} \cdot R = \frac{190}{235} \cdot 200 = 161.70\,\Omega/\square$$

Alternatively, expressed in terms of resistance per area:

$$R_\square = \frac{W}{L \cdot R} = \frac{190}{200 \cdot 235} \approx 0.00404\,\Omega/\mu\text{m}^2 = 4.04\,\text{k}\Omega/\text{m}^2$$

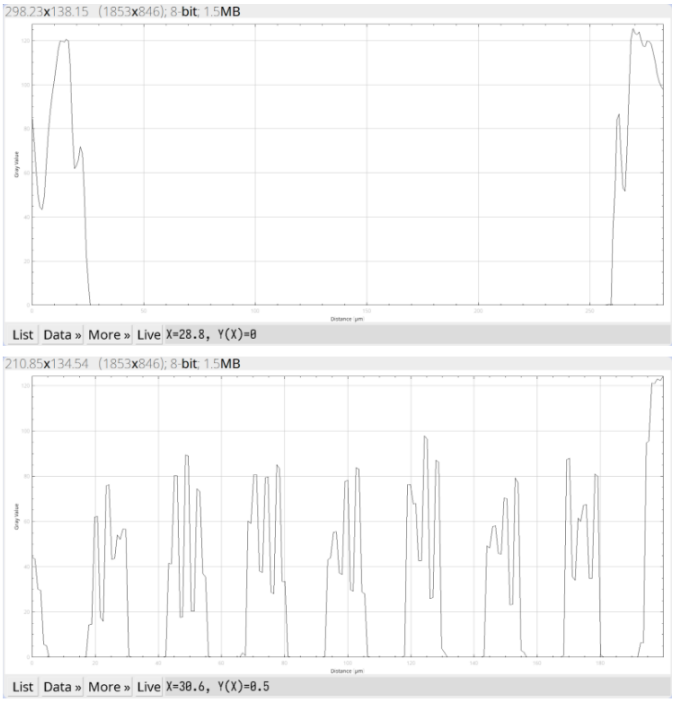


Fig. 14. ImageJ Measurements for the Heating Resistor

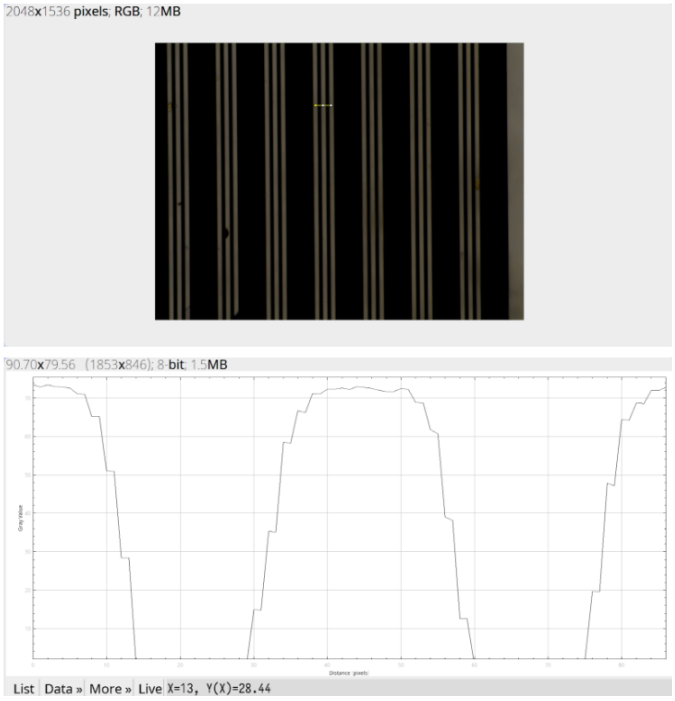


Fig. 15. Measuring Resistor ImageJ Measurement

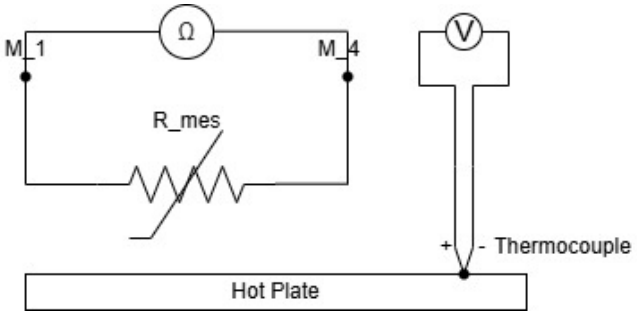


Fig. 16. Experimental Setup Explicit Diagram

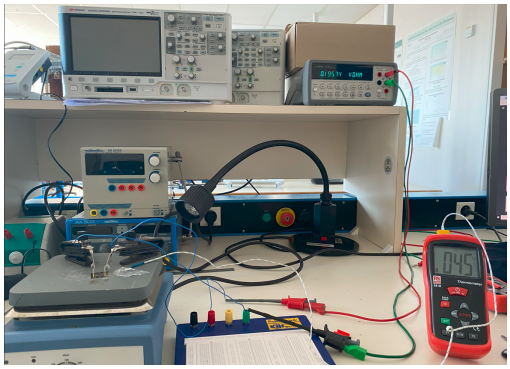


Fig. 17. Laboratory Experimental Setup for Temperature Measurement Resistor

B. Characterisation of the Temperature Measurement Resistor

Thermo-cycler functions as a control mechanism for temperature. It needs to have an adjust system and a measuring system such that the instantaneous temperature data can be known and changed if needed. The measuring resistor is a thermistor in which the change in temperature results in a variation in resistance value. If the proportional relation is known, the measured resistance can give the temperature. To utilise the thermo-cycler produced in the microfabrication steps, a characterisation setup is prepared that can be seen in the figure 2. A thermocouple and hot plate allows the control of temperature while an ohmmeter is connected to the measuring resistor to record the data. Thermocouple is connected to the hot plate with an aluminum tape for conduction and secure positioning. The recorded measuring resistance values are given in table II below.

These data are plotted and a best-fit graph is obtained in MATLAB through polyfit function, as can be seen in figure 3.

From this graph, we can deduce that there is a linear relationship between the temperature and the measuring resistance. Thus, we can obtain the formula for the best-fit graph that can be used while employing the thermocycler in the PCR application. The linear-fit equation is as follows:

$$R_{\text{mes}} = 0.55\,T + 174.82$$
$$R_{\text{mes}} = 174.82\,(1 + 0.00315\,T)$$

TABLE III
MEASURED RESISTANCE R_m AT DIFFERENT TEMPERATURES

Temperature (°C)	**Measured R_m (Ω)**
22	188.0
27	191.0
40	193.7
45	195.7
55	206.0
65	214.0
75	218.0
85	222.0
95	227.0
105	236.0
115	238.0
120	238.0

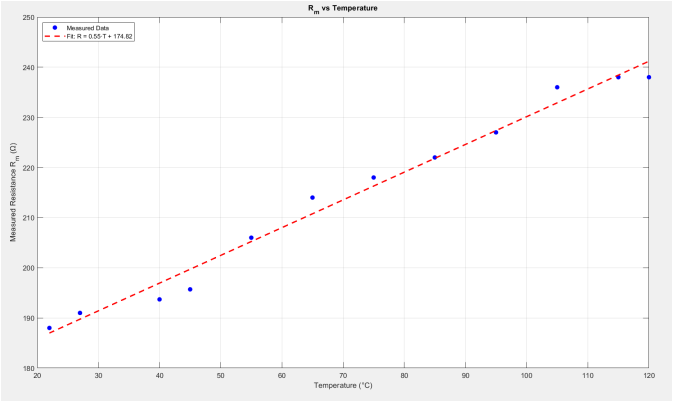


Fig. 18. Measuring Resistance and Temperature Graph with Polyfit in MATLAB

This equation allows us to approximate the parameters in the linear temperature coefficient of resistance model.

$$R_m(T) = R_{\text{mo}} * (1 + (\alpha m)T)$$

In this equation, αm is the estimated thermal resistance coefficient. In the literature, a common temperature probe such as the Pt100 sensor is reported to have a thermal coefficient of approximately 0.00385 Ω/C [3]. Even though the values deviate, the lower thermal coefficient observed in our sample suggests reduced sensitivity to temperature changes, which may be advantageous in applications requiring thermal stability or where minimal variation is desirable over changing thermal conditions.

C. Characterisation of the Heating Element

In this section, heating resistor which dissipates power to increase temperature is characterised. Voltage source is chosen to be the independent variable, while the current flowing through the heating resistor is recorded as well. Voltage source enables current to flow throught the heating resistor which in turn dissipates energy in the form of heat. We can measure the temperature with the previous characterisation of the measuring resistance.

The recorded data can be seen in the table III. While continuous measurement is also made for the current value

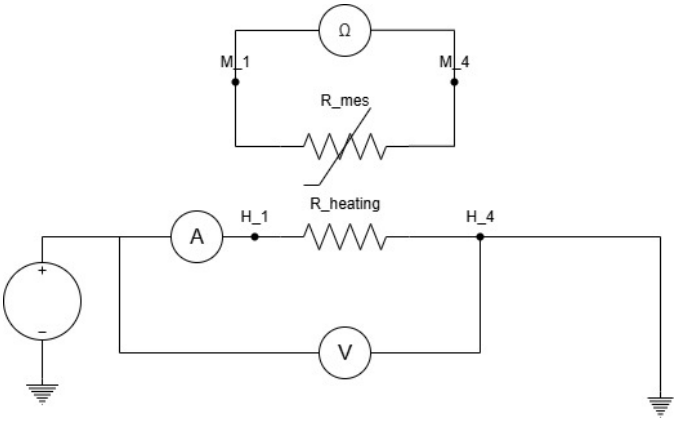


Fig. 19. Circuit Explicit Diagram



Fig. 20. Laboratory Experimental Setup for Heating Element Characterisation

for the calculation of electrical power. The relation between the voltage and measuring resistance can be seen in the plot where a best-fit curve is drawn to approximate certain non-empirical voltage values. Where the best-fit function is given below:

$$R = 0.0024 \cdot V^2 + 0.0082 \cdot V + 0.1797$$

In order to examine the transfer of electrical power, the voltage values and the continuously measured current values are multiplied with respect to the equation below and another plot is drawn with a best-fit function:

$$P = V * I$$

The best-fit function for the power and resistance graph is given below:

$$R = -0.0272 \cdot P^2 + 0.2023 \cdot P + 0.1842$$

Through the polyval function in the MATLAB, both best-fit graphs resulted in second-degree polynomials. However, upon

TABLE IV
MEASURED VOLTAGE, CURRENT, AND RESISTANCE VALUES

Voltage (V)	Current (mA)	Resistance (kΩ)
0.282	7.81	0.184
0.661	18.13	0.186
0.947	25.70	0.188
1.298	33.10	0.19599
1.517	40.00	0.19599
1.844	47.70	0.20000
2.120	53.90	0.20626
2.440	60.60	0.21775
2.980	71.00	0.22460
3.320	77.10	0.23320
3.570	80.70	0.23950
3.880	85.50	0.24794
4.250	90.40	0.25750
4.530	94.30	0.26730
4.820	98.00	0.27323
5.020	100.40	0.27920

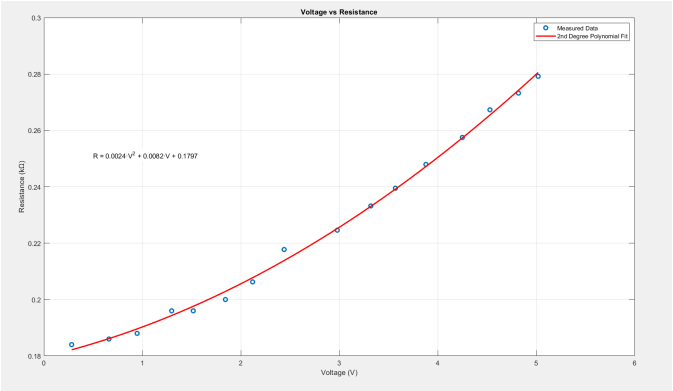


Fig. 21. Voltage vs. Measurement Resistance with Best-Fit Curve

closer inspection, in terms of the power and resistance graph, the linear terms' coefficient dominates the quadratic term. Hence, linear approximation can be sensible in low power region, such that the increasing quadratic term's effect can be ignored. As in the characterisation, voltage is limited to the 0 - 5V range that allows this approximation. This allows the control of micro-heating for the PCR application such that

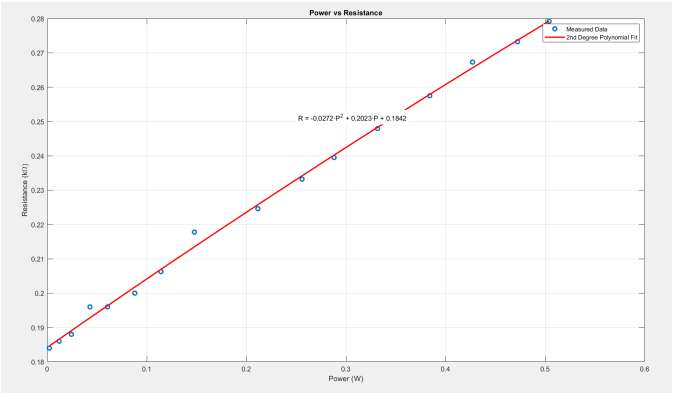


Fig. 22. Electrical Power vs. Measuring Resistance with Best-Fit Curve

through this function, accurate heating can be achieved. Previous characteristics function for the temperature and measuring resistance is required for this controlling mechanism.

D. Balance Sheet and Operations

1) Objective: The objective of this section is to formulate a mathematical model that determines the link between the electrical power (P) and the subsequent temperature (T) of the PCR microchip. This relationship is important for achieving the correct thermal cycling needed for the operation of PCR, where the power input determines the temperature of the chip.

Two equations from the previous characterisation steps were obtained such that they related temperature and power with resistance.To model the thermal behavior of the chip, resistance (R) was algebraically eliminated between these two equations to find a direct relationship between Power (P) and Temperature (T).

2) Derivation: By equating the two expressions of resistance, we obtain:

$$T(P) = \frac{-0.0272 \times P^2 + 0.2023 \times P + 0.1842 - 174.82}{0.55}$$

3) Curve Fitting and Interpretation : Despite the quadratic relationship, for low power (where most micro-PCR applications fall), the behavior of the overall function is almost linear in nature. This makes for a more straightforward approximation, which is more practical for embedded systems that require real-time thermal control. This model is essential for integrating a heating control logic into the chip's use environment.

4) Comparison to Measuring Resistor Model : The previously used temperature estimation model via measuring resistor:

$$R_{\text{mes}} = 0.55\,T + 174.82$$

is consistent with the calculated thermal-power model. The measuring resistor provides a direct measurement of temperature while the effective equation uses a predictive model from power put into the resistor which allows closed loop temperature control even in the absence of sensor feedback, if it cannot be immediately determined.

5) Practical Application: How to Control Temperature for PCR : To achieve PCR thermal cycling, power is sent to the heating resistor based upon what stage of the PCR protocol the system is trying to achieve (denaturation, annealing, extension). The microcontroller can use the above equation to determine the necessary power input to achieve the target temperature. Real-time readings from the measuring resistor can be used for feedback and correction to stay at the target temperature.

$$T \approx 336.68 \cdot P + 19.15$$

The observed linear relation between resistance and temperature is in line with the standard model for metallic materials:

$$R = R_0(1 + \alpha T)$$

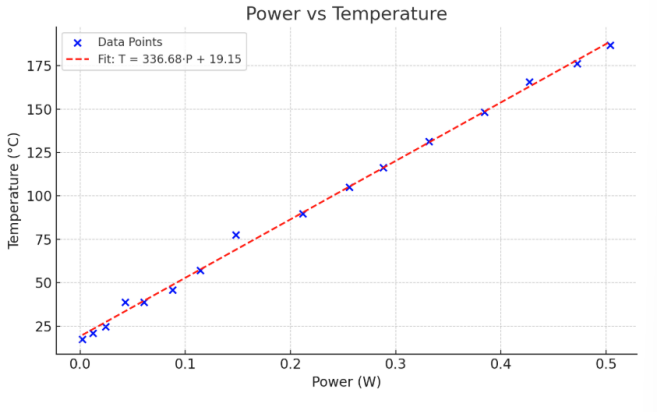


Fig. 23. Electrical Power vs. Temperature with Best-Fit Curve

The equation relating temperature and power indicates that temperature increases proportionally with the supplied electrical power, with a sensitivity of approximately 336.68 °C/W

This model was proven valid for a range of experiments of something like 0 to 0.6 W, where the relationship of power to temperature is fairly linear. The more accurate and efficient use of the applied power, in turn, will mean that temperature can be more accurately controlled as a function of applied power. This method of temperature control is especially relevant for PCR, which relies on changing from temperature to temperature in a relatively linear, smooth, rapid, and reliable fashion.

6) Temperature Estimation Method Comparison : In this section, two techniques were tested to estimate the PCR chip temperature:

1. Resistance Temperature Model Here, the initial relationship between resistance and temperature of the sensing resistor is used. This method can give a high level of accuracy, as the sensing resistor was characterized under controlled conditions, using a thermocouple to spatially map the chip temperature accurately.

2. Power Temperature Estimation By discarding the intermediate resistance values using power vs resistance plot, a direct relationship between electrical power and temperature was created. This model could be used when dynamic temperature control is needed since the power input can also be controlled in real time, and therefore it can provide fast estimates of temperature without having to add additional measurement circuitry.

The resistance-based model is ideal for precise calibration and validation. The power-based model is advantageous for real-time temperature control in PCR cycling.

According to third characterisation section, at 2.98 V, the resistance was measured as 224.60 , which matches the result predicted by the resistance-temperature calibration. This contract establishes a level of confidence in the models we developed.

7) Temperature Control Strategy in Chip PCR Cycles : The microfabricated PCR-chip must control a precise and programmable temperature to carry out a PCR cycle. One PCR

cycle will consist of three surgical heated temperatures:

Denaturation (94–98°C)

Annealing (50–65°C)

Extension (72°C)

These thermal stages have been accurately implemented on-chip for control purposes. The heating-resistant (R_{heater}) is applied at a voltage/current to increase temperature, and the measuring-resistor (R_{mes}) is utilized as a real-time thermal sensor, as its resistance decreases linearly with increasing temperatures.

Closed-loop temperature control can be implemented using a feedback system:

1) Power Input: Voltage is adjusted to deliver electrical power to the heater resistor (P = V·I).
2) Real-Time Monitoring: The resistance of Rmes is continuously measured. Using the previously calibrated R–T relationship (Characterisation 2), the system estimates the temperature.
3) Feedback Algorithm**:** A microcontroller (for instance, a PID controller) compares the estimated temperature with the desired setpoint. If the temperature is either too low or too high, it adjusts the voltage up or down, respectively.

Optional Enhancement: Adding a digital control system (e.g., microcontroller with ADCs) could enable the system to cycle itself through PCR stages. To enhance accuracy, it could employ real-time curve fitting or a look-up table based upon the R–T data obtained experimentally.

Conclusion: The chip has the capabilities to execute multiple rapid PCR cycles with accurate timing and temperature control through this feedback-based thermal regulation system making portable and energy-efficient DNA amplification on a microscale platform feasible.

E. Dynamic Study

In this section, the aim is to obtain the response time of the circuit. In order to achieve this, a 3V step voltage input is applied to the heating resistor. By controlling this variable, then the measuring resistor is observed through a constant current source and an oscilloscope. The reasoning is to measure the resistance by recording voltage values such that through Ohm's Law, the resistance will be the division of the voltage with constant current. As the constant current source, Howland source is constructed with LF356 OPAMP. The explicit diagram can be seen below.

The experiment results can be seen in the oscilloscope result in fig(), Primarily, the initial voltage is measured to be 189.55 mV and this value can be converted to resistance as shown below:

$$V = I \cdot R \Rightarrow \frac{V(mV)}{I(mA)} = R(\Omega)$$

$$I = \frac{5V}{4.7k\Omega} = 1.06mA \approx 1mA$$

Hence the initial measuring resistance is 189.55 Ω while the final resistance value can be observed as 225.73 Ω. When

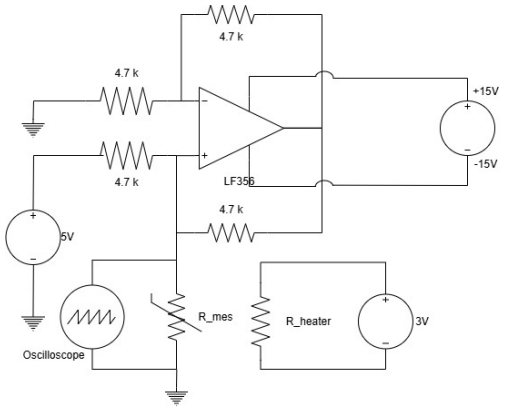


Fig. 24. Circuit Explicit Diagram for Dynamic Study Characterisation

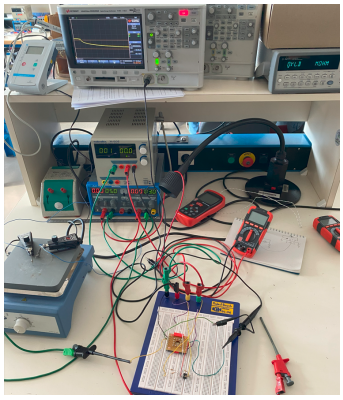


Fig. 25. Laboratory Experimental Setup for Dynamic Study Characterisation



Fig. 26. Heating Response in Measuring Resistance



Fig. 27. Cooling Response in Measurement Resistor

compared to the characterisation of heating element, 2.980 V value is measured to result in 224.60 Ω in table III. Hence, through the endpoints, the dynamic study is validated. Upon inspection, the response aligns with a first-order circuit step response. Steinmetz states that the thermal model can be described through the same form of first-order step response of an RC circuit [4]. Also, as in our circuit, there is no initial state, the response is solely zero-state response in heating stage. Thus, the thermal output in the oscilloscope can be described with the equation below:

$$V(t) = V_f \cdot (1 - \exp(\frac{-t}{\tau}))$$

To determine the heating time constant, the equation is utilized when t = τ:

$$V_t = V_f \cdot (1 - e^{-1}) = v_t = 0.632 \cdot V_f$$

$$0.632 \cdot (225.73mV - 189.55mV) + 189.55mV = 212.42mV$$

With the help of grid intervals, the time when the voltage equates to 212 mV can be approximately measured to be 1.25 seconds. The time interval is 1 seconds and voltage interval is 20 mV as shown in the oscilloscope screen.

$$\tau_H = 1.25s$$

The same analysis is conducted for the cooling case as well. The oscilloscope result for the cooling curve is given below:

Due to cursor misalignment, the initial and final voltage values are modified by inspection of the grid intervals.

$$V_i = 229.73mV$$

$$V_f = 207.55mV$$

In cooling case, we can also use the first-order step response formula, however this case have the dominant zero-input response term as zero-input response is ignored. The equation can be given as below:

$$V(t) = V_i \cdot (\exp(\frac{-t}{\tau}))$$

To determine the time constant, same procedure is followed with the variation such that 0.367 of the initial voltage to final voltage drop is used to pinpoint the t = τ result. 215.72 mV is found to be the value after τ seconds pass. Hence, the cooling time constant is found to be:

$$\tau_C = 0.5s$$

It is found that the system exhibits different time constants during heating and cooling. The heating phase shows a time constant of approximately 1.5 seconds, while the cooling phase is significantly faster, with a time constant of about 0.5 seconds. This difference indicates that the system heats up more slowly than it cools down. This can be explained such that cooling benefits from heat dissipation to the environment. Hence, it is faster than the heating.

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